BA ZC581 :
Applied Data Science
Module: 5

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Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 5: Business Databases with Python

Module 5 Topics

Topic 1: Introduction to Databases & SQL

Topic 2: NoSQL using Python (Mongo, PostgreSQL)

Topic 3: Demos of Data IO Operations

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Topic 1: Introduction to Databases & SQL

Introduction to Databases

- What is a Database?
  - A centralized data store for enterprise data that powers all applications
  - An organized collection of data, stored and accessed electronically.
  - Designed to manage large volumes of information efficiently.
- Why Databases Matter
  - Enable storage, retrieval, and manipulation of data.
  - Support consistency, security, and scalability in applications.
  - Power everything from websites to banking systems.

Types of Databases

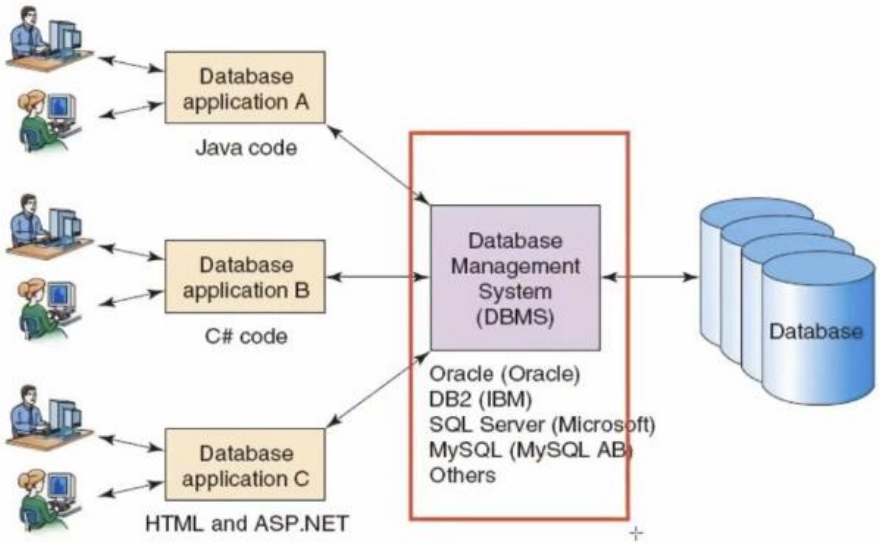
- Relational (SQL): structured, table-based (e.g., MySQL, PostgreSQL)
- NoSQL: flexible, schema-less (e.g., MongoDB, Cassandra)
- Other Types
  - OO Databases
  - XML Databases
  - Multimedia Databases
  - Graph Databases
  - Geospatial Databases
  - Time Series Databases

DBMS/ RDBMS

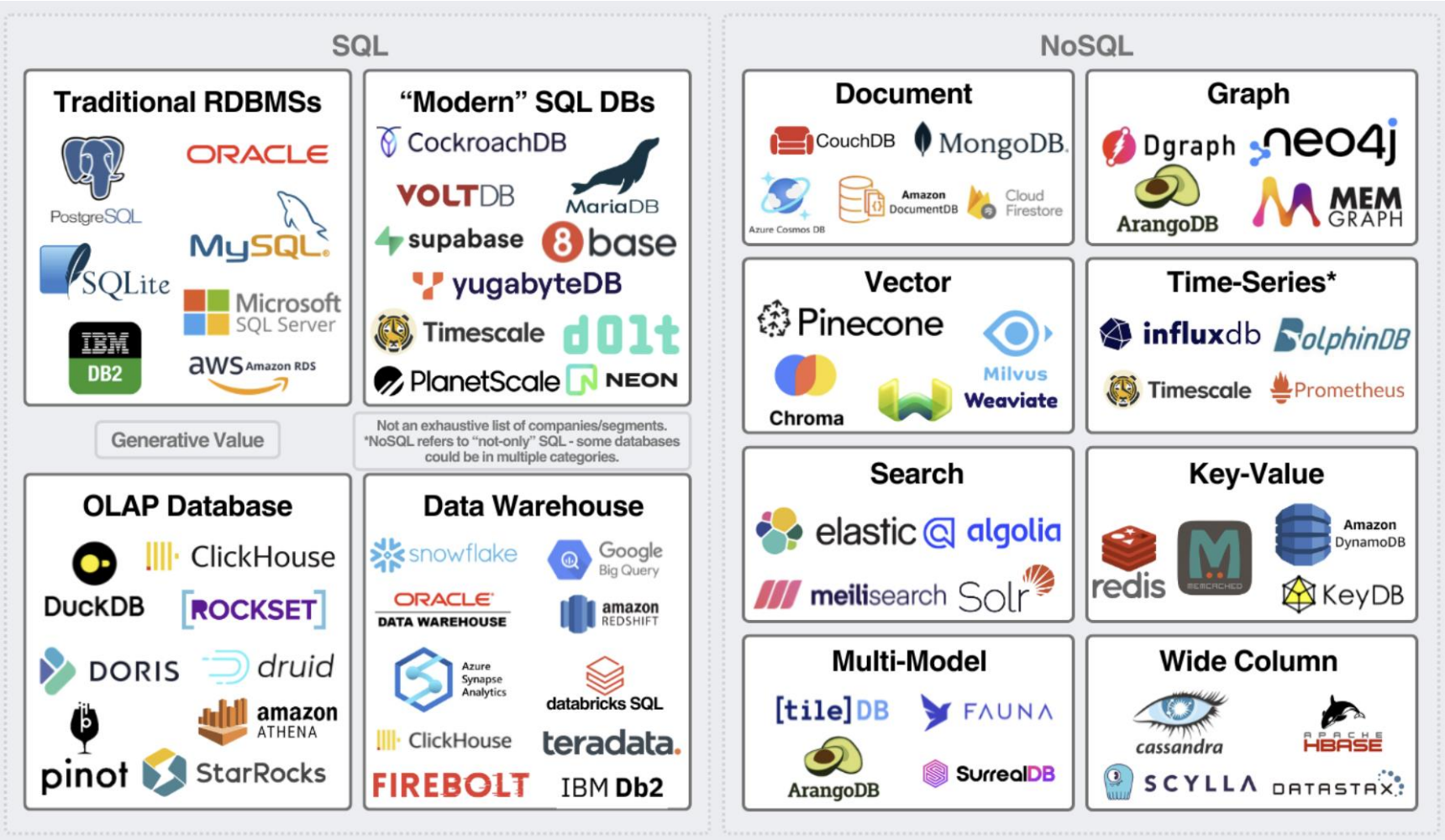
- Data Base Management System (DBMS) is the software program to store, retrieve, manage user's data stored in a 'database'.
- **RDBMS (Relational DBMS)**
  - Software to manage data stored in related tables (table: rows and columns of data)
  - Allows operations to create, read, update, insert, delete data from the database (CRUD operations)
  - Provides mechanism to define the structure of the data and also administer it.
  - **Uses SQL (Structured Query Language) as a standard language to perform operations**
  - Ex: Oracle, MSSQL Server, MySQL, DB2

The RDBMS

- RDBMS is a Client- Server program that manages the data in the data storage and also provides mechanism for clients to access/operate on data.
- Client gives a request for data / operation to be performed; and the RDBMS server carries out the operation to produce the result data.
- Offers Data Access Services
- Uses SQL language



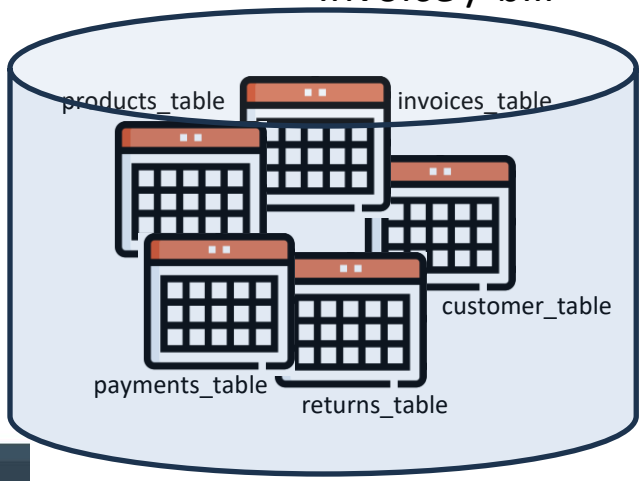
Database Landscape



A Small Shop Scenario

products

Invoice / bill



Sales Database

customers

payments

returns

Customer service

Logical Entities of a Table

Employee Data in Employee Table:

Columns stores a specific data type

Primary Key (Int)

Fields

Row
Or record

Emp No	Name VarChar	Age Int	Department VarChar	Salary Amount
001	Alex S	26	Store	5000
002	Golith K	32	Marketing	5600
003	Rabin R	31	Marketing	5600
004	Jons	26	Security	5100

Table : Employee

A column whose values uniquely identifies a record is called Primary Key

Relational Structure

- Most widely used structure
  - Data elements are stored in tables
  - Row represents a record; column is a field
  - Can relate data in one table with data in another, if both tables share a common data element

Department Table

Deptno	Dname	Dloc	Dmgr
Dept A	Design	Pune	
Dept B	Sales	Chennai	
Dept C	Finance	Mumbai	

Employee Table

Empno	Ename	Etitle	Esalary	Deptno
Emp 1	Amit	Manager	30,00,000	Dept A
Emp 2	Radha	Officer	12,00,000	Dept A
Emp 3	Bipin	Manager	26,00,000	Dept B
Emp 4	Robin	Officer	12,00,000	Dept B
Emp 5	Amit	Officer	17,00,000	Dept C
Emp 6	Ryan	Steno	8,00,000	Dept B

Relation: employee working in a department

Schema

- The logical structure of how the database is structures and how the tables relate to each other.
- Typically, data is split into multiple interrelated tables that allows fragments of data to be ‘picked’ and assembled from multiple ‘related’ tables

The Data Schema

Entities:

- User
- Order
- LineItem
- Product

LineItem relates

- Order & Product

User

UserID	User	Address	Phone	Email	Alternate
1	Alice	123 Foo St.	12345678	alice@example.org	alice@neo4j.org
2	Bob	456 Bar Ave.		bob@example.org	
...	...	...	...	...	...
99	Zach	99 South St.		zach@example.org	

Order

OrderID	UserID
1234	1
5678	1
...	...
5588	99

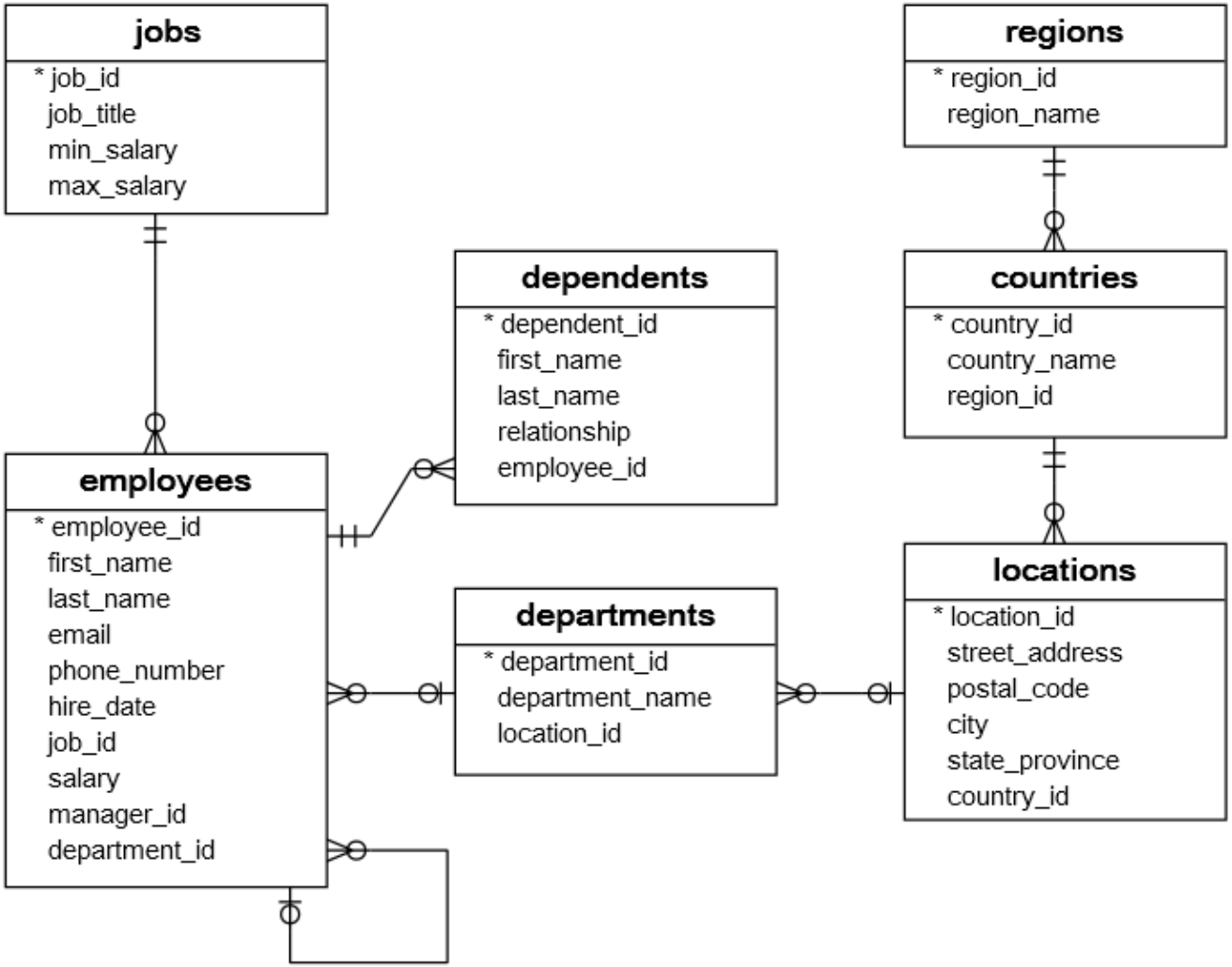
LineItem

OrderID	ProductID	Quantity
1234	765	2
1234	987	1
...	...	...
5588	765	1

Product

ProductID	Description	Handling
321	strawberry ice cream	freezer
765	potatoes	
...	...	
987	dried spaghetti	

Example 2: Employee DB Schema



Tech Jargon

- **Query**: A command that performs a single operation on the data. like insert, update, select or delete operation.
- **SQL**: A standard rules (and a programming language syntax) that are used to query a relational database.
  - Databases that allow SQL way of access to data are called SQL databases. (Most RDBMS are SQL compliant)
- **Relational DB**: Data is organized as a table and allows data to be accessed in relation to another table's columnar data
- **NoSQL**: NoSQL refers to a non relational database where data is stored without strict schema; data is stored as key-value pairs; documents in semi structured form.
- **Document oriented DB**: A database that stores document information (semi structured form)
- **DBMS**: A software that manages / administers the data and offers a controlled access to the data (CRUD operations)

Tech Jargon, contd

- **Transaction**: A sequence of operations treated as a single unit of work. It either completes fully (commit) or has no effect at all (rollback). This ensures the database remains consistent, reliable, and correct even in the face of errors or failures.
- Database transactions follow the **ACID** properties:
  - **Atomicity**: Every transaction is unique and performs an atomic unit of work. All operations succeed or none do.
  - **Consistency**: The database moves from one valid state to another, respecting all rules and constraints.
  - **Isolation**: Transactions run independently without interfering with each other, even if executed concurrently.
  - **Durability**: Once committed, changes are permanent, even if the system crashes.

Database Operations

- **C**reate:
  - Create anew
  - Insert / add new data
- **R**etrieve / Read:
  - Retrieve
- **U**pdate:
  - Read – Modify – Write back. Change the existing data
- **D**elete:
  - Remove

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Structured Query Language & MS SQL Server

Structured Query Language (SQL)

- **Structured Query Language**
  - Used to manipulate the datab
  - Simple command like statem
  - Standard and independent o
- Data Definition Language St
  - Used to create the structure of tab the overall schema
- Data Manipulation Language Statements
  - Used to perform operations on stored data

SQL: Structured Query Language, used by most RDBMS

SQL Operations on Database

DATA MANIPULATION

- **SELECT** – extracts data from a database
- **UPDATE** – updates data in a database
- **INSERT** – inserts new data into the database
- **DELETE** – deletes data from database

DATA DEFINITION

- **CREATE** – creates a new database / table
- **ALTER** – modifies the database structure
- **DROP** – deletes the table data and structure

Database Interrogation

- SQL Queries
  - Query format is SELECT…FROM…WHERE…

Natural Language:

Prepare a Department wise list of Employees whose salary is between 450000 and 750000

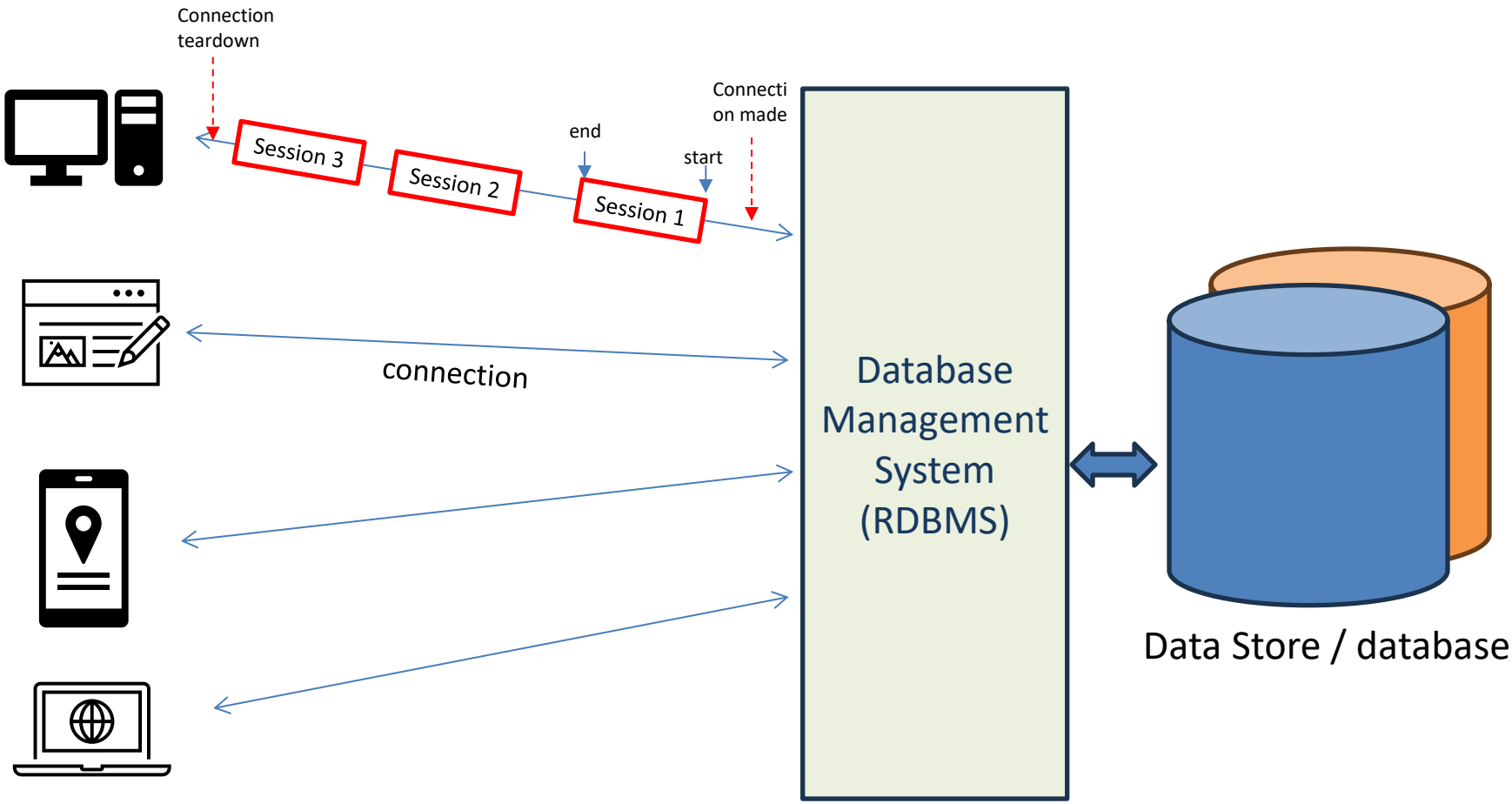
SQL:

```
SELECT  EMP_ID,  EMP_NAME,  EMP_SALARY
   FROM EMPLOYEES
   WHERE  EMP_SALARY  BETWEEN  450000  AND 750000
   GROUP BY DEPT_NAME
   ORDER BY EMP_NAME
```

Connection Mechanism

- User Application (clients) establishes a connection to the database.
  - uses a socket mode of networked connection on a specific port.
  - user us authenticated (authorization is done)
- User application sets up a session
- Operations are carried out as part of the session
- Session can close; and reopen any number of times with the same connection; (time out)
- Connection is closed when DB services are no longer needed

Connection Mechanism



connection string: Driver={PostgreSQL};Server=IP address;Port=5432; Database=myDataBase; Uid=myUsername; Pwd=myPassword;

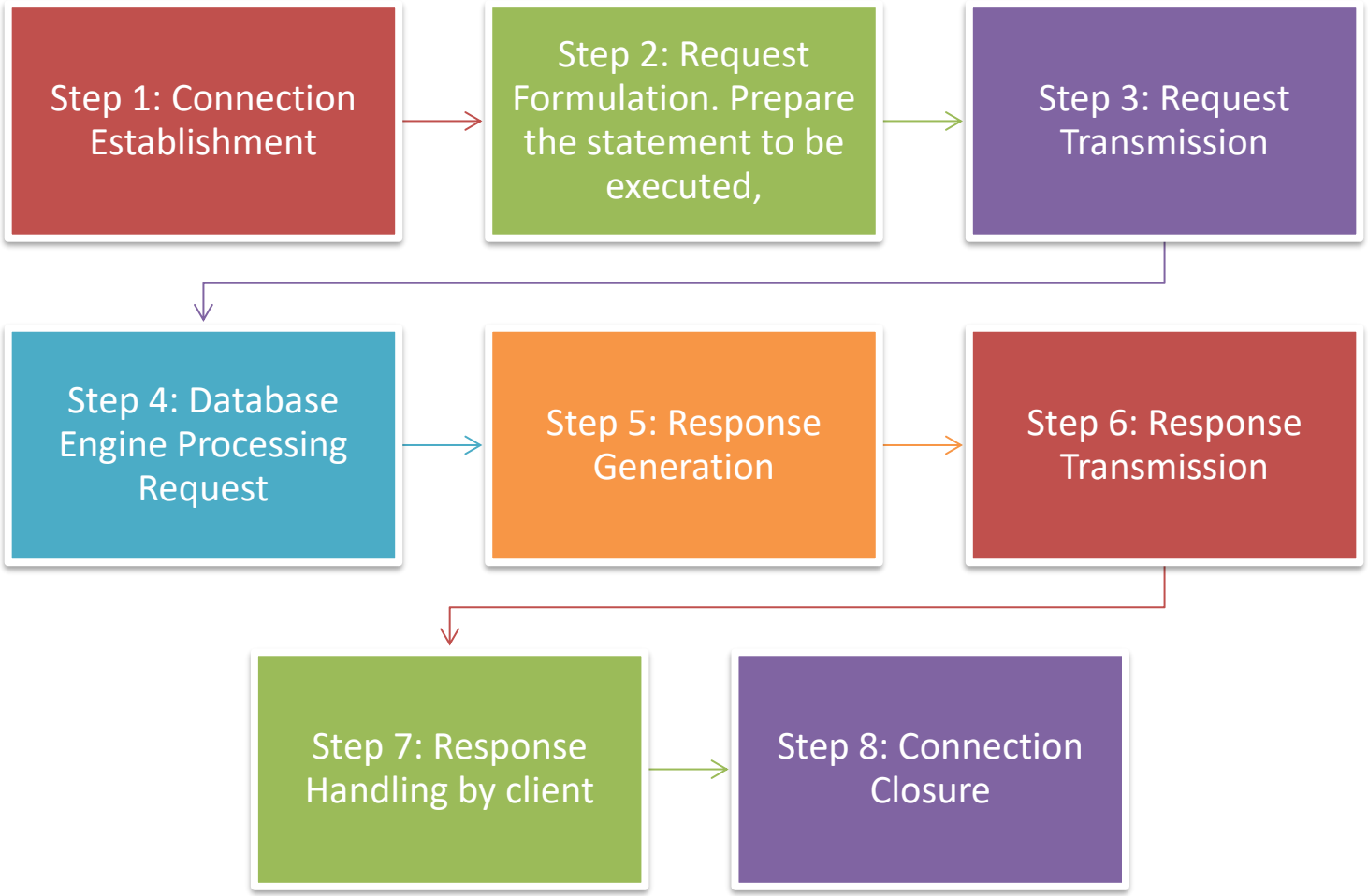
The Using Connection String (odbc)

ADO.NET
SQLAlchemy
Apache Sqoop
Python DB-API
MuleSoft DB Conn

- odbc = Open Data Base Connector
- jdbc = Java Data Base Connector
- Ways to Connect
  - Use a client Program, mostly a GUI based program used to Manage / Administer / Operate the database
  - Use a power shell to run queries
  - Use a dbshell to connect and run queries
  - Use a programming language like Java, C++, Python to connect and run queries to operate and access data.
- Connection String:
  - Combination of Server name, DB Instance name, User name, password

connection string: Driver={PostgreSQL};Server=IP address;Port=5432; Database=myDataBase; Uid=myUsername; Pwd=myPassword;

DB Access from a program



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MS SQL Server

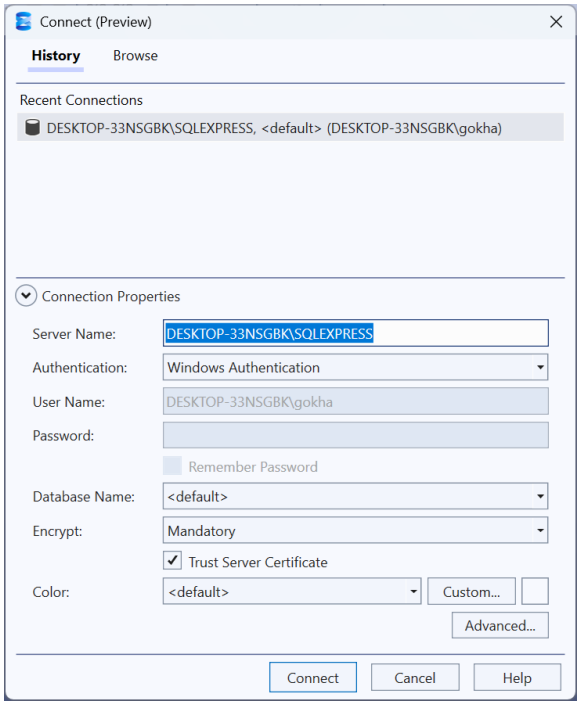
(Express Edition)

MS SQL Server

- Microsoft SQL Server is a RDBMS that supports a wide variety of transaction processing, business intelligence (BI) and data analytics applications in corporate IT environments.
- SQL Compliant
- MS SQL Server Express Edition – free to download and use
- Supports
  - Indexing
  - Stored Procedures
  - Transactions
  - Buffering
  - Connection Pooling and More
- Server Management Studio is the Admin / Management console

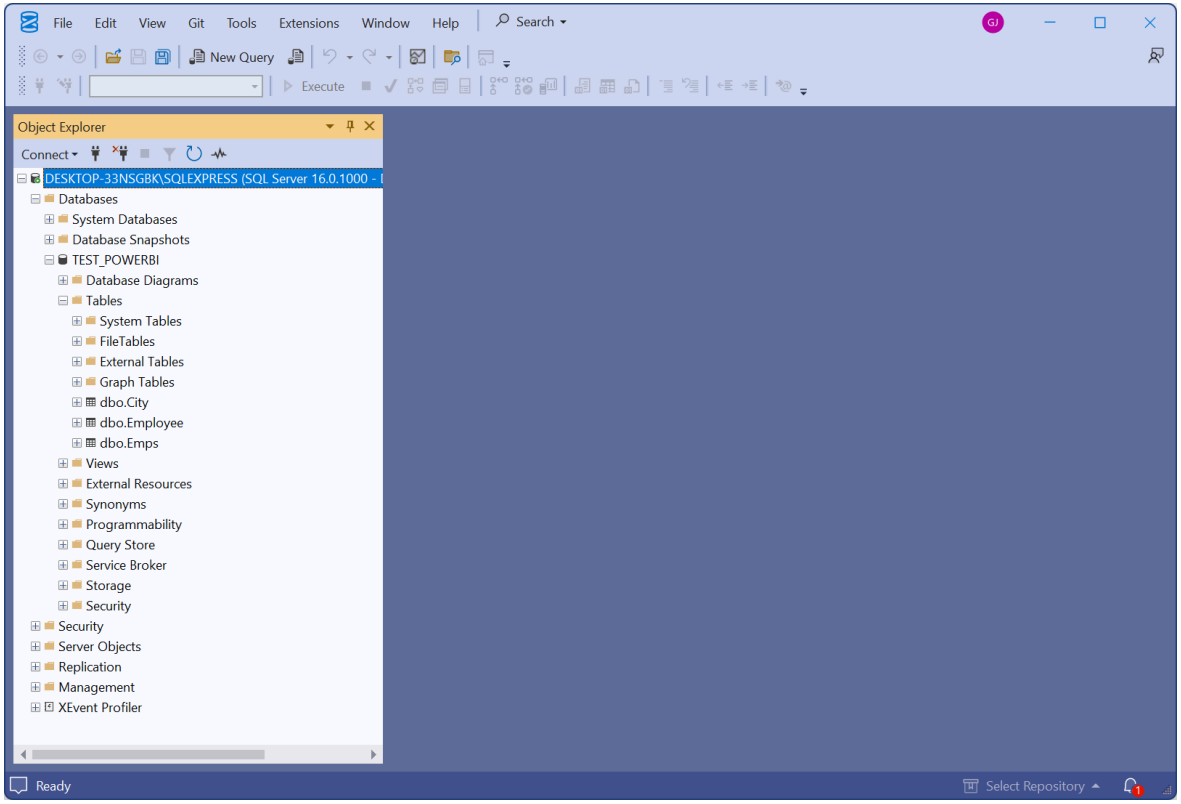
Connect to MS SQL Server EE

- Connect to the MS SQL Server instance and launch the Server
- Connect to the Server from a client application to perform operations

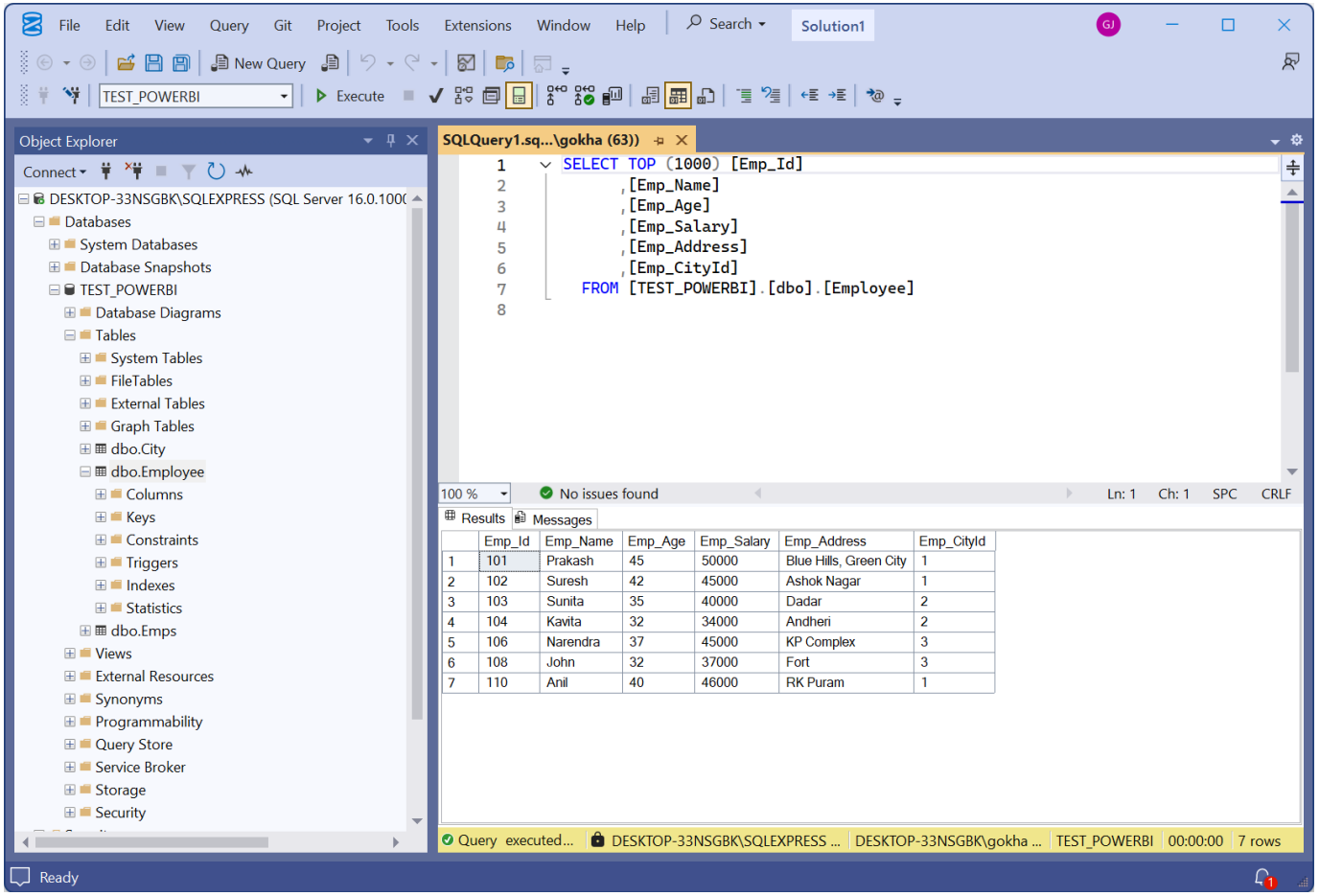


SQL EE Server Management Studio

The Admin Console for MS SQL Server EE



SQL Server in Action



Connect with Python

Connect to Emps table and access data.

cs5 demo programs for database access\001-db access with mssql.py

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Topic 2: NoSQL using Python (PostgreSQL and MongoDB)

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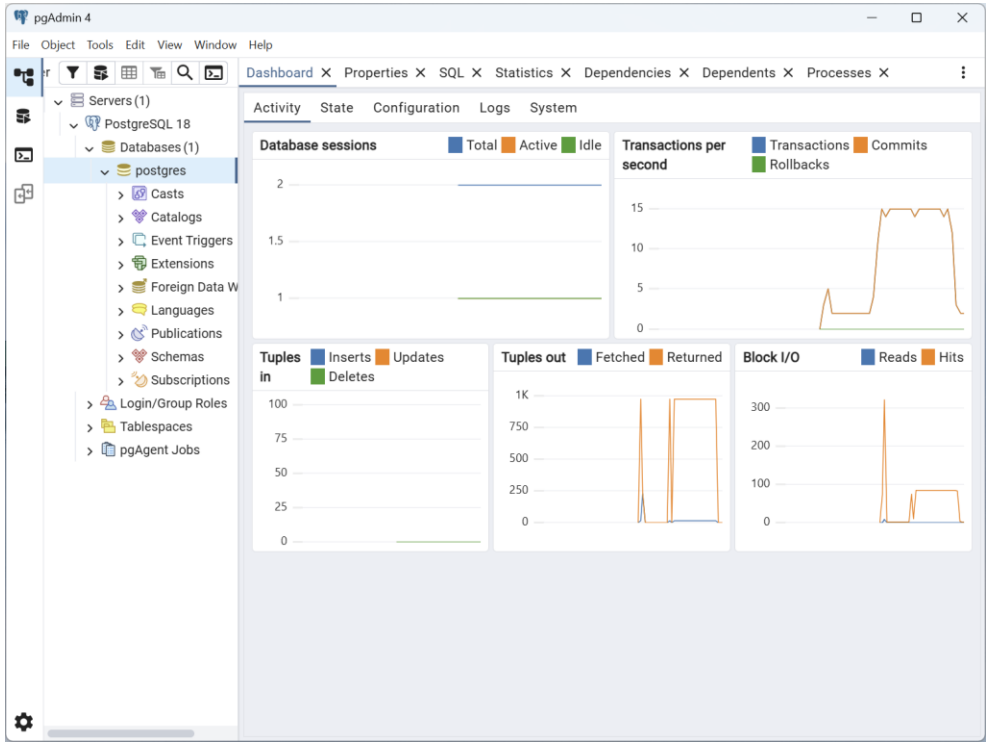
PostgreSQL DB

PostgreSQL and pgAdmin

- PostgreSQL
  - Open-source, advanced relational database system
  - A RDBMS that also supports non relational data types like JSON, hence also called Object Relational DB (ORDBMS)
  - Supports standard SQL
  - Supports complex queries, transactions, and ACID compliance
  - Extensible with custom data types, functions, and indexing
  - Ideal for enterprise apps, analytics, and web backends
- pgAdmin
  - Official GUI management tool for PostgreSQL
  - Provides visual dashboards for queries, schemas, and performance monitoring
  - Simplifies database administration (backup, restore, user management)
  - Cross-platform and widely used by developers & DBAs
- Use Together
  - PostgreSQL = powerful database engine
  - pgAdmin = intuitive interface for managing PostgreSQL
  - Together: efficient, scalable, and user-friendly database management

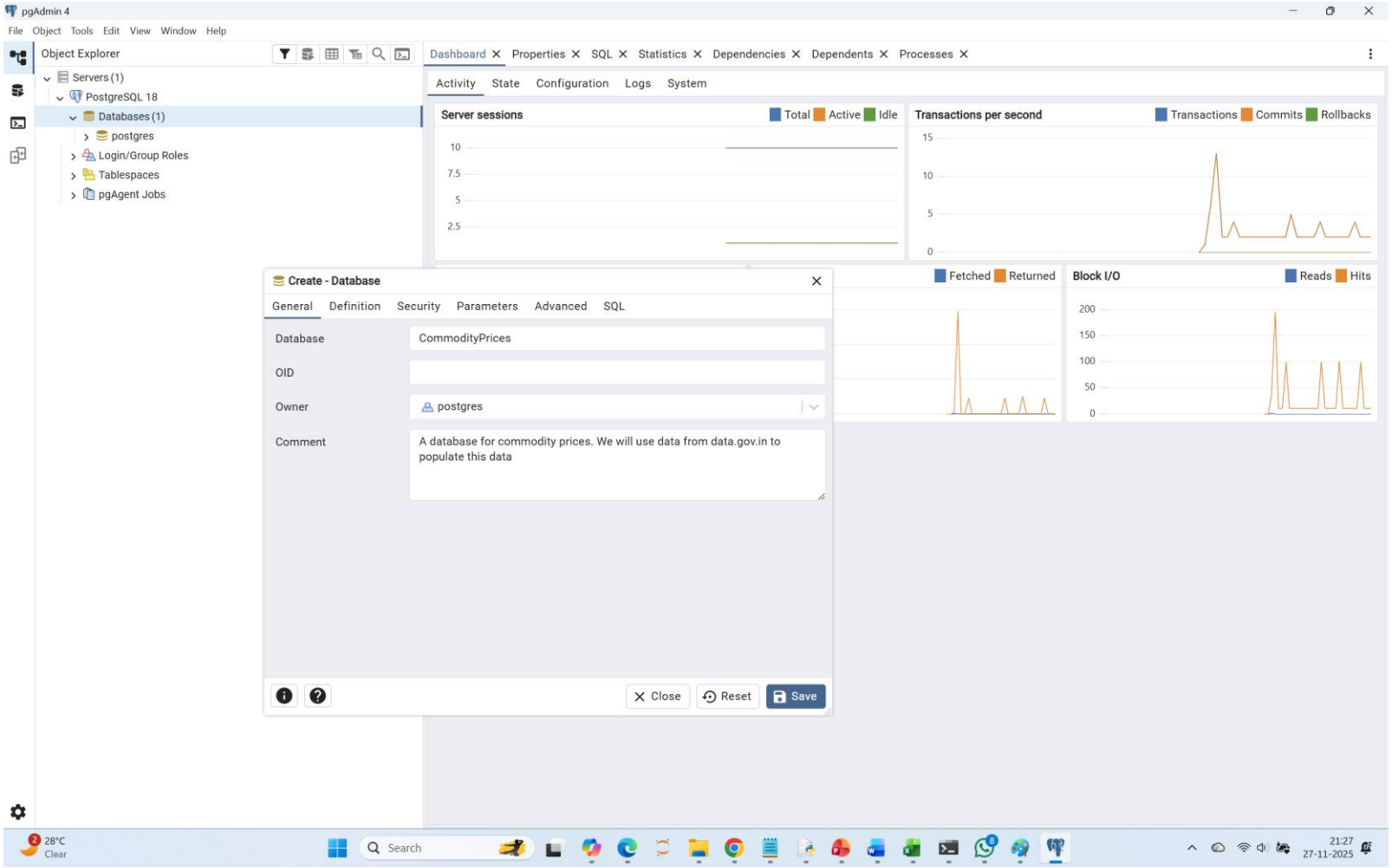
pgAdmin

A console to access, administer and operate PostgreSQL database

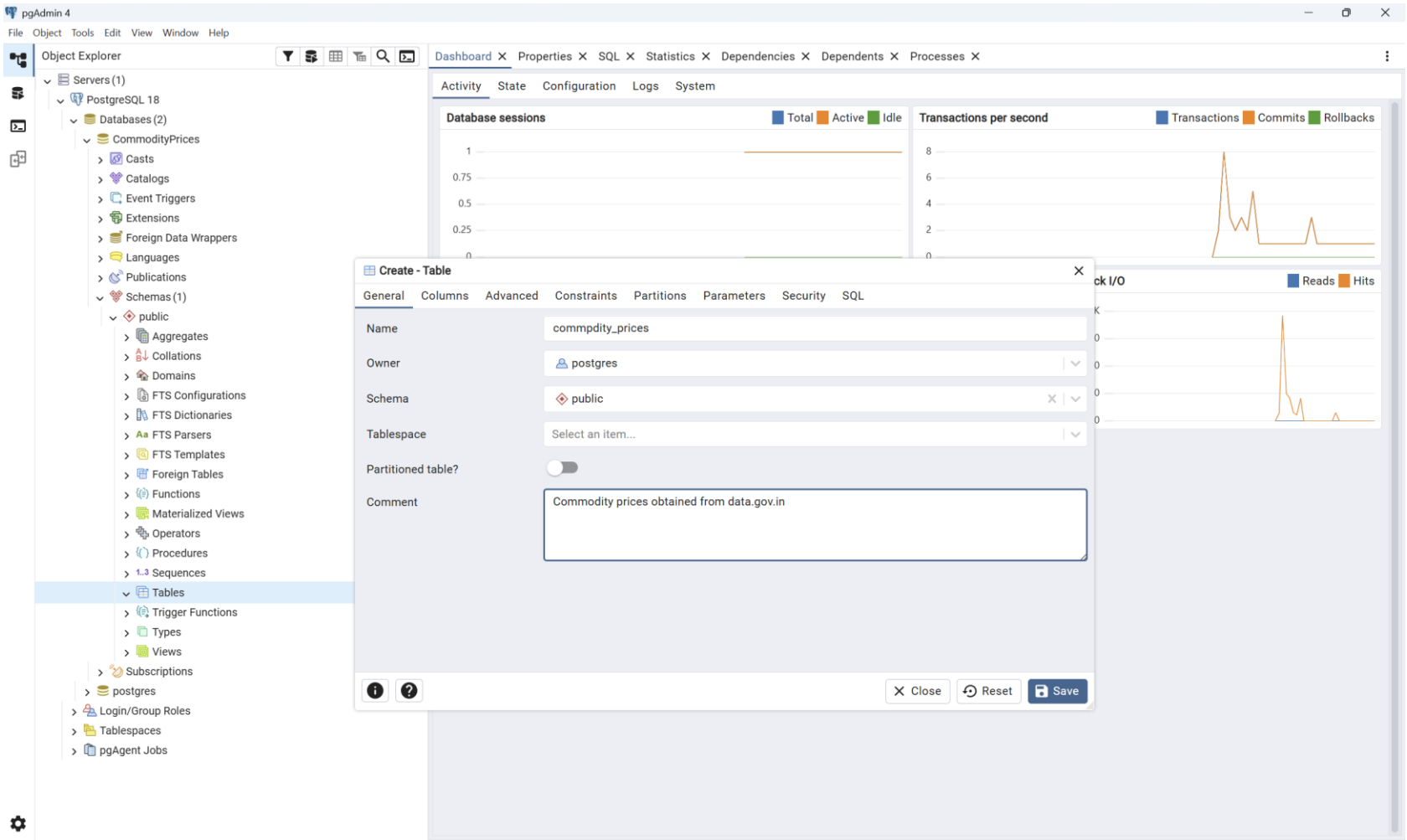


Install pgAdmin from: www.pgadmin.org/download

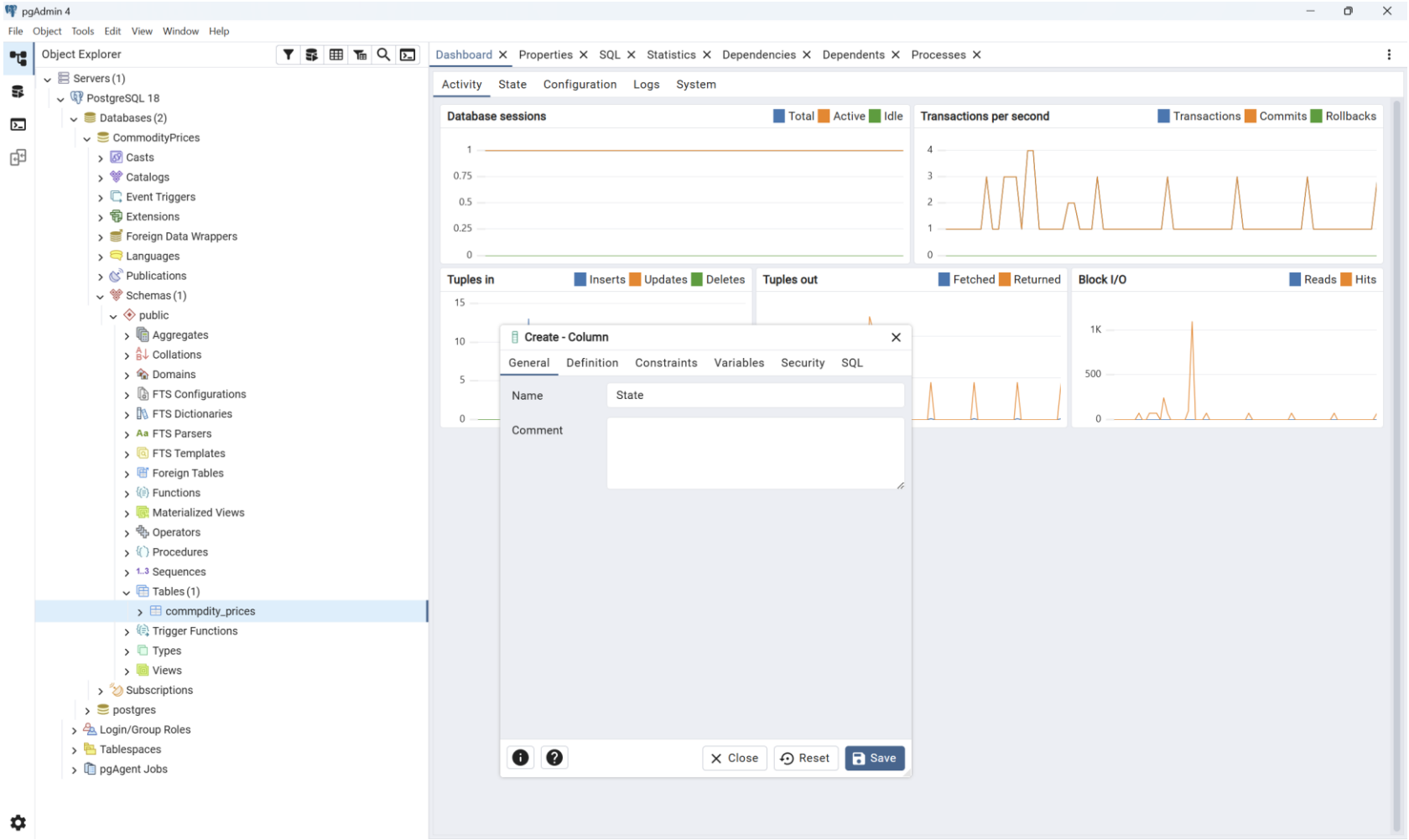
Create Database



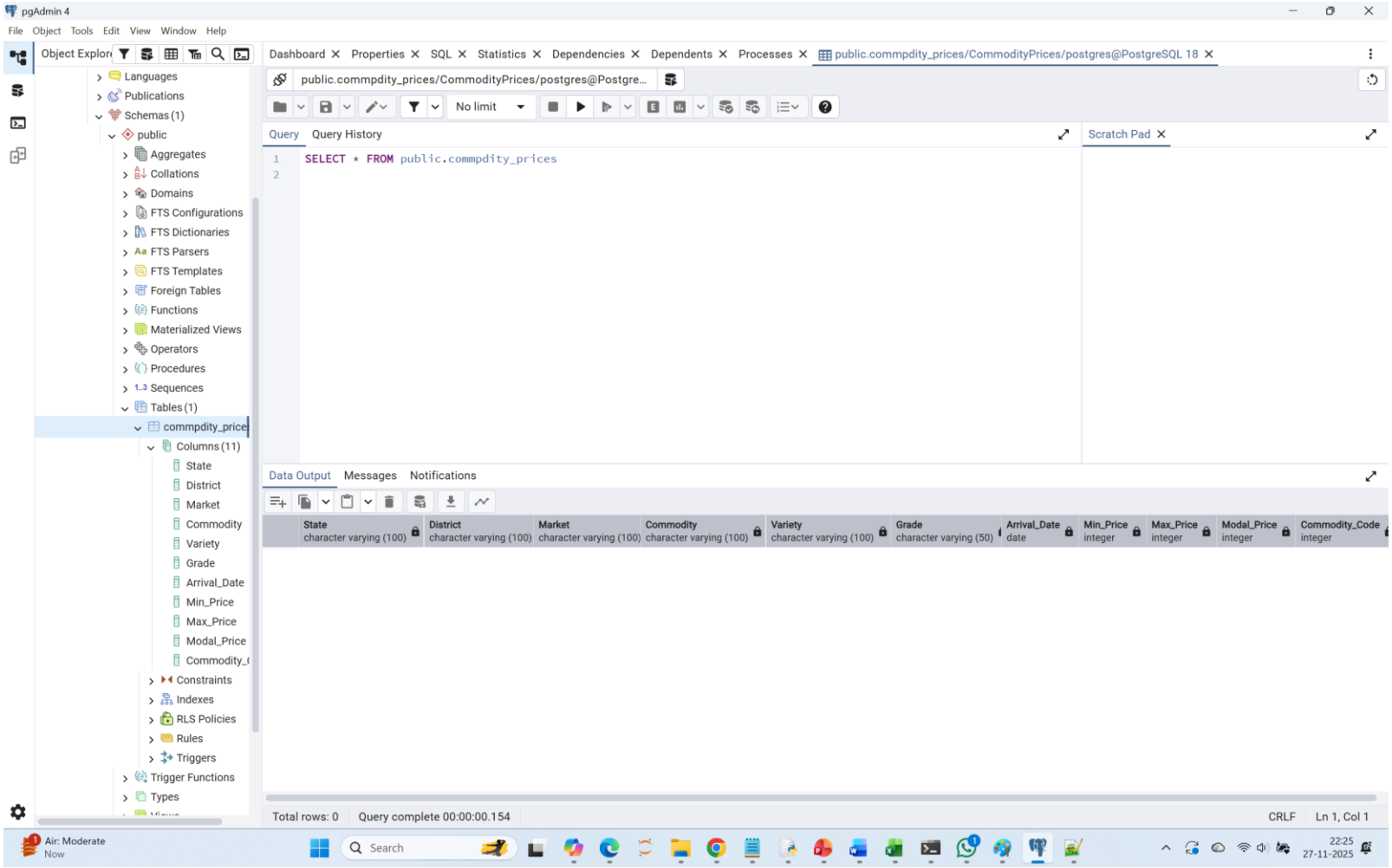
Create Table



Create Column

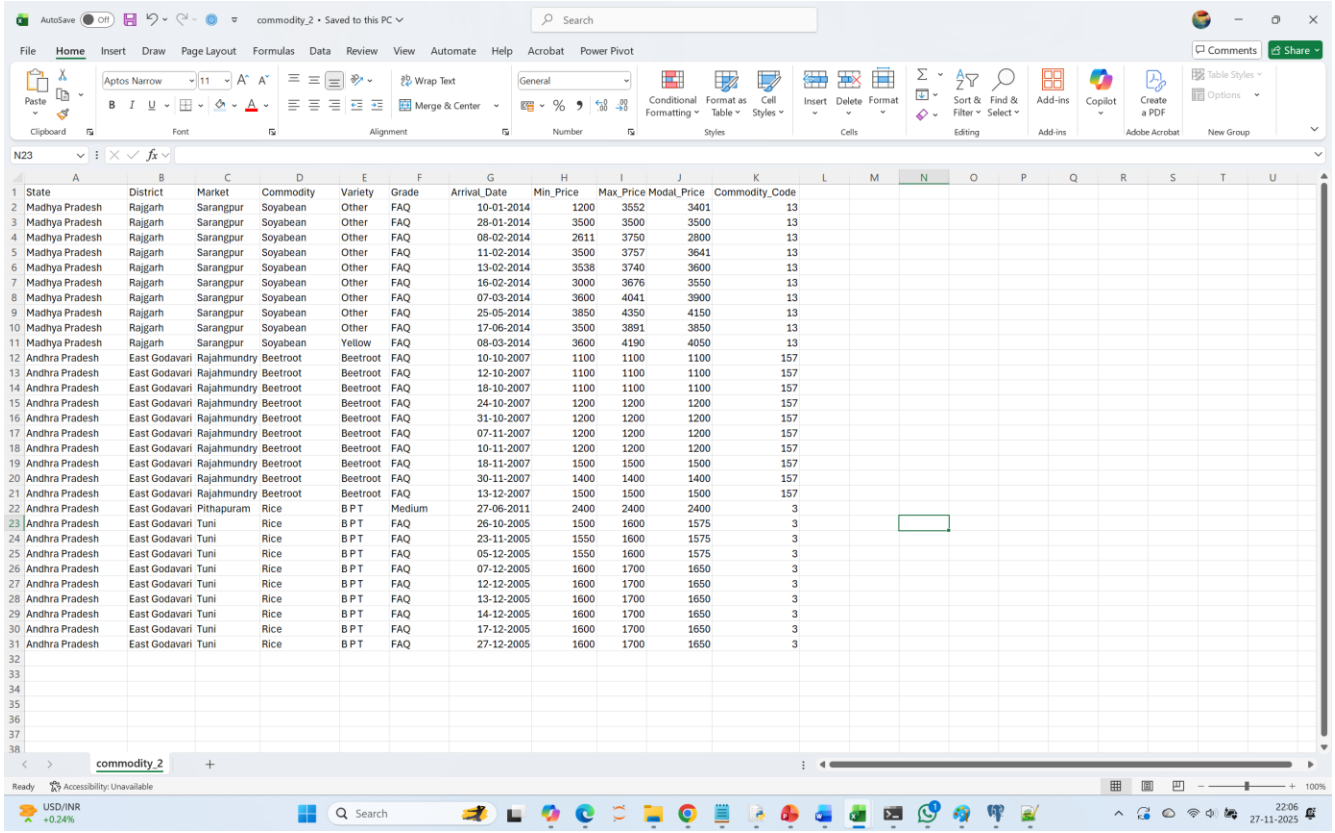


Create Columns as per the table structure



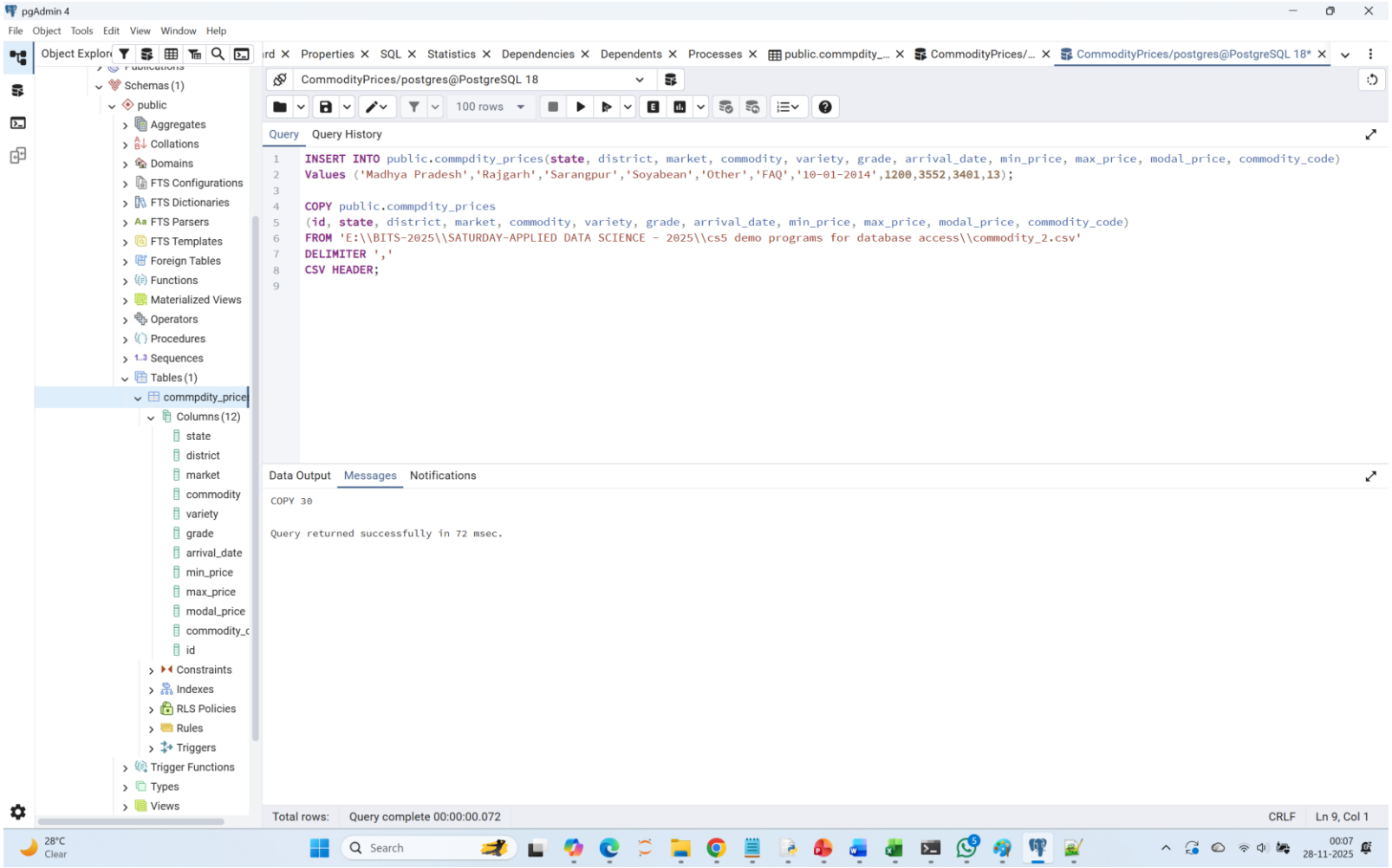
Data extracted from data.gov.in website

30 records extracted using public GUI. Using the API, we can extract data as well. Data is saved as commodity_2.csv

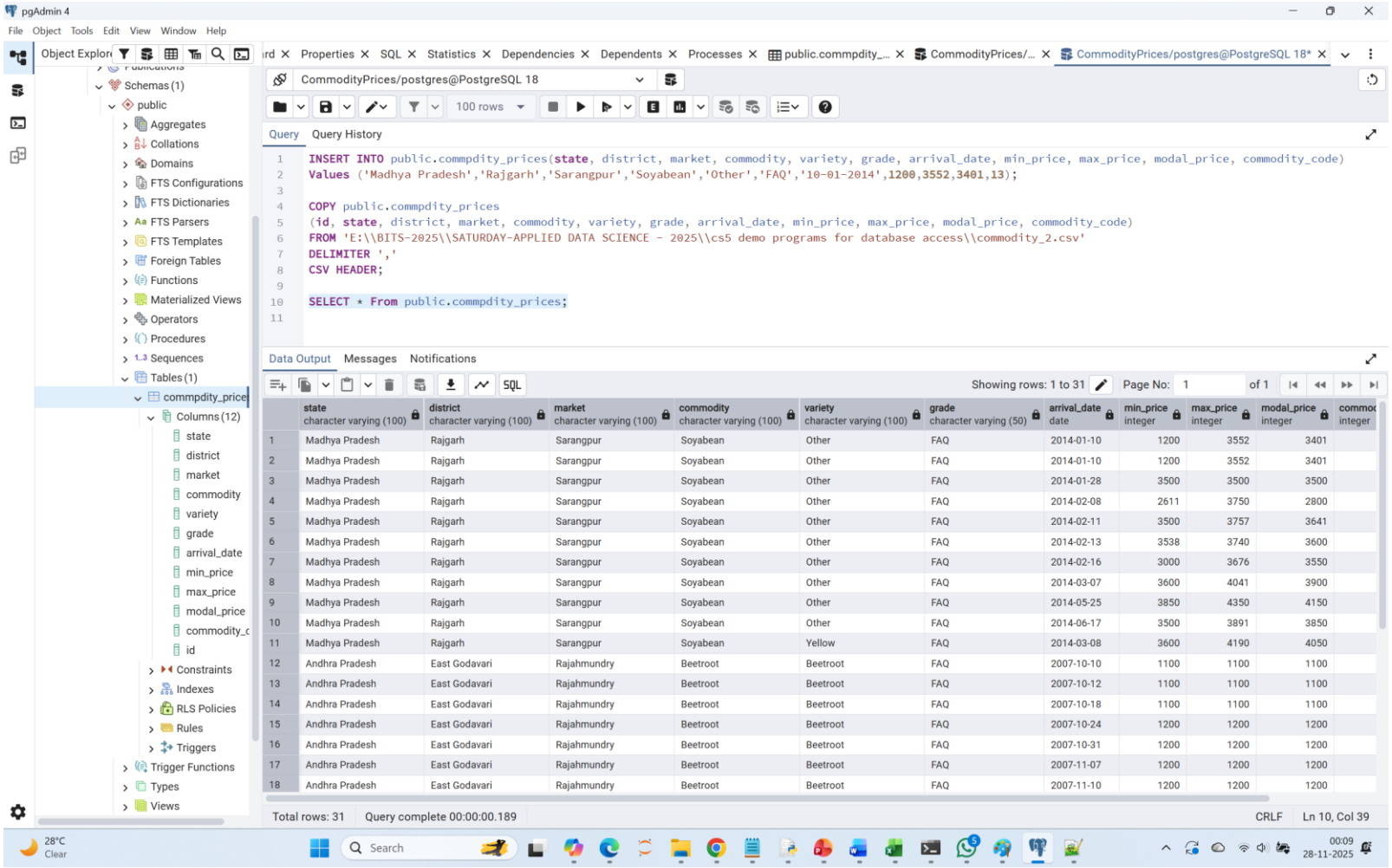


State	District	Market	Commodity	Variety	Grade	Arrival_Date	Min_Price	Max_Price	Modal_Price	Commodity_Code
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	10-01-2014	1200	3552	3401	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	28-01-2014	3500	3500	3500	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	08-02-2014	2611	3750	2800	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	11-02-2014	3500	3757	3641	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	13-02-2014	3538	3740	3600	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	16-02-2014	3000	3676	3550	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	07-03-2014	3600	4041	3900	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	25-05-2014	3850	4350	4150	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	17-06-2014	3500	3891	3850	13
Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Yellow	FAQ	08-03-2014	3600	4190	4050	13
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	10-10-2007	1100	1100	1100	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	12-10-2007	1100	1100	1100	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	18-10-2007	1100	1100	1100	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	24-10-2007	1200	1200	1200	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	31-10-2007	1200	1200	1200	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	07-11-2007	1200	1200	1200	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	10-11-2007	1200	1200	1200	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	18-11-2007	1500	1500	1500	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	30-11-2007	1400	1400	1400	157
Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	13-12-2007	1500	1500	1500	157
Andhra Pradesh	East Godavari	Pithapuram	Rice	B P T	Medium	27-06-2011	2400	2400	2400	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	26-10-2005	1500	1600	1575	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	23-11-2005	1550	1600	1575	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	05-12-2005	1550	1600	1575	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	07-12-2005	1600	1700	1650	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	12-12-2005	1600	1700	1650	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	13-12-2005	1600	1700	1650	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	14-12-2005	1600	1700	1650	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	17-12-2005	1600	1700	1650	3
Andhra Pradesh	East Godavari	Tuni	Rice	B P T	FAQ	27-12-2005	1600	1700	1650	3

Insert / import Data from csv



View Data using Select



	state character varying (100)	district character varying (100)	market character varying (100)	commodity character varying (100)	variety character varying (100)	grade character varying (50)	arrival_date date	min_price integer	max_price integer	modal_price integer	commoc integer
1	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-01-10	1200	3552	3401	
2	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-01-10	1200	3552	3401	
3	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-01-28	3500	3500	3500	
4	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-02-08	2611	3750	2800	
5	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-02-11	3500	3757	3641	
6	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-02-13	3538	3740	3600	
7	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-02-16	3000	3676	3550	
8	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-03-07	3600	4041	3900	
9	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-05-25	3850	4350	4150	
10	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Other	FAQ	2014-06-17	3500	3891	3850	
11	Madhya Pradesh	Rajgarh	Sarangpur	Soyabean	Yellow	FAQ	2014-03-08	3600	4190	4050	
12	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-10-10	1100	1100	1100	
13	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-10-12	1100	1100	1100	
14	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-10-18	1100	1100	1100	
15	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-10-24	1200	1200	1200	
16	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-10-31	1200	1200	1200	
17	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-11-07	1200	1200	1200	
18	Andhra Pradesh	East Godavari	Rajahmundry	Beetroot	Beetroot	FAQ	2007-11-10	1200	1200	1200	

Python access to PostgreSQL

Demo Program to display records from commpdity_prices :

cs5 demo programs for database access\001-db-access with postgressql.py

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NoSQL Databases & Mongo DB

NoSQL Databases

Definition

- Non-relational databases designed for flexible, scalable data storage.
- Handle unstructured, semi-structured, and structured data.

Key Features

- Schema-less: No fixed table structure.
- High scalability: Horizontal scaling across servers.
- Flexible data models: Document, key-value, column, graph.
- Fast performance: Optimized for big data and real-time apps.

Types of NoSQL Databases

- Document-based (MongoDB, CouchDB)
- Key-Value stores (Redis, DynamoDB)
- Column-family (Cassandra, HBase)
- Graph databases (Neo4j, OrientDB)

Use Cases

- Real-time analytics
- IoT and sensor data
- Social media apps
- E-commerce product catalog

Types of NoSQL DB

Key-value stores: Data is stored as a collection of key-value pairs, making them very simple and efficient for use cases like shopping carts or user profiles.

Document databases: Data is stored in "documents," which are self-contained units of data, often in formats like JSON or XML. This is ideal for content management and cataloging.

Wide-column stores: Data is stored in tables with columns, but unlike relational databases, the columns can be different for each row. This is useful for handling massive datasets with varying structures.

Graph databases: Data is stored as nodes, edges, and properties, allowing for the representation and traversal of complex relationships between data points. This is used for social networks or recommendation engines

SQL & NoSQL

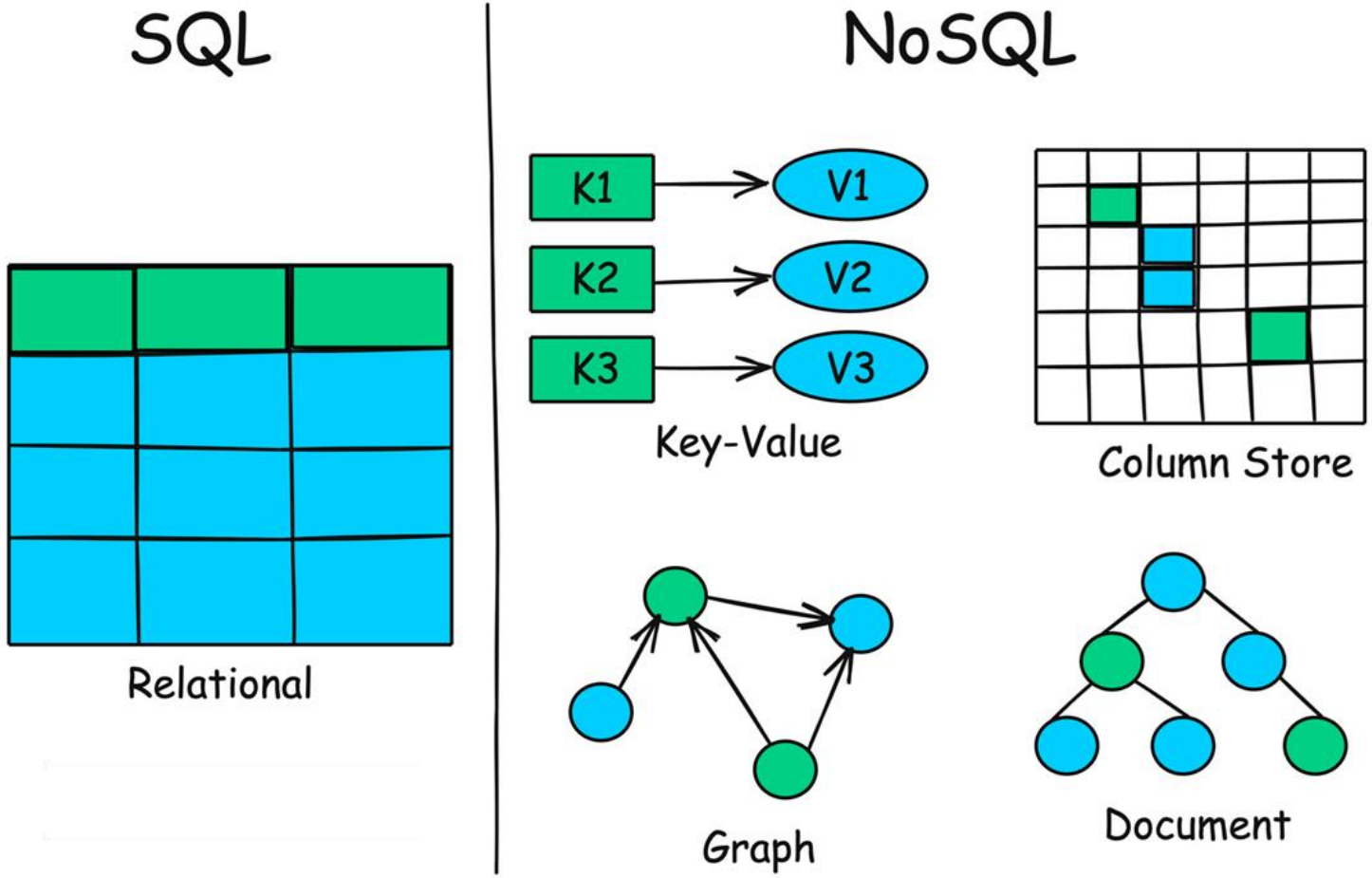


Image courtesy: algomaster.io

SQL vs NoSQL: A Comparison

Aspect	SQL	NoSQL
Data Model	Relational, table-based with fixed schema	Various models (key-value, document, column-family, graph) with flexible schema
Scalability	Vertical scaling (adding more power to existing machines)	Horizontal scaling (adding more machines to the pool)
Schema	Rigid, predefined schema	Dynamic, flexible schema
Query Language	SQL (Structured Query Language)	Varies by database, often lacks a standard query language
Transactions	Supports ACID properties (Atomicity, Consistency, Isolation, Durability)	Varies; often supports BASE properties (Basically Available, Soft state, Eventual consistency)
Consistency	Strong consistency	Eventual consistency (in some cases)
Joins	Supports complex joins	Limited or no support for joins
Use Case	Suitable for applications with complex queries and transactions	Ideal for big data, real-time analytics, and flexible data models
Examples	MySQL, PostgreSQL, SQL Server	MongoDB, Cassandra, Redis, Couchbase
Typical Applications	Financial systems, ERP, CRM	Social networks, content management, IoT

MongoDB Community Edition & Compass Client

- MongoDB is a NoSQL Document Oriented Database.
- Community Edition is completely free to use for learning, development, and even production workloads.
- Document-Oriented Model: Stores data in flexible, JSON-like documents instead of rigid tables.
- Features Included:
  - Ad-hoc queries
  - Secondary indexing
  - Real-time aggregations
  - Schema-less design (data can evolve without strict schema changes)
- Supported Platforms: Available for Windows, macOS, and Linux.
- Ideal for students, startups, and developers who want to experiment or build applications without licensing fees
- MongoDB Atlas is a fully managed cloud service

Key Concepts of MongoDB

Documents

- The basic unit of data in MongoDB.
- Stored in BSON (Binary JSON), which extends JSON with additional data types like dates and binary.
- Documents can have nested fields and arrays, making them highly flexible.

Collections

- A collection is a group of documents, similar to a table in relational databases.
- Collections do not enforce a strict schema unless you configure them.
- Documents in the same collection can have different fields

Key concepts, contd.

Databases

- A database is a container for collections.
- Each MongoDB instance can host multiple databases.
- Useful for logically grouping related collections.

ObjectId

- Each document has a unique identifier called ObjectId.
- It's a 12-byte hexadecimal value that ensures uniqueness across distributed systems

Indexes

- Indexes improve query performance.
- MongoDB supports single-field, compound, text, and geospatial indexes.
- Without indexes, queries must scan the entire collection

Key concepts, contd..

Replica Sets

- A replica set is a group of MongoDB servers that maintain the same data.
- Provides high availability and automatic failover.
- One node is primary (handles writes), others are secondary (replicas).

Sharding

- Distributes data across multiple servers (horizontal scaling).
- Useful for handling massive datasets and high-throughput workloads.
- Data is partitioned using a shard key.

MongoDB

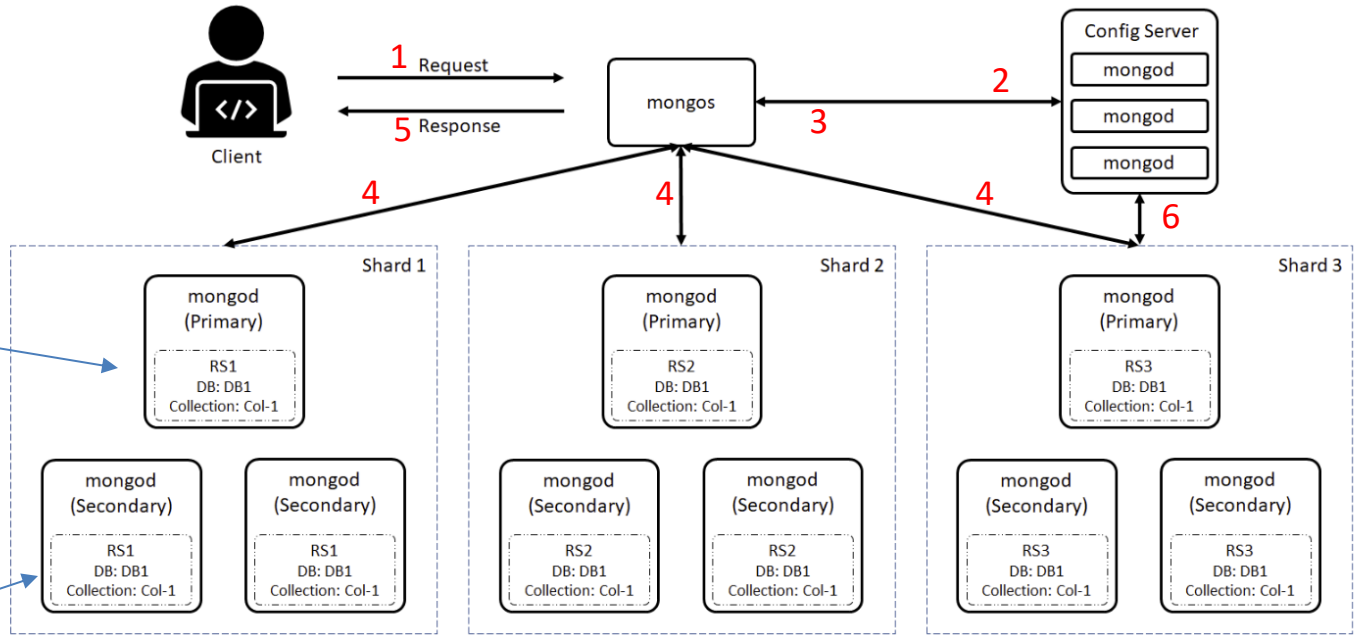
mongos: Mongo Shard router

mongod: MongoDB daemon; the main db manager

Config Server: holds meta data of every chunk in the shard.

performs writes

performs reads / replicates / takes control as master if primary master fails



MongoDB Architecture

MongoDB basic Operations

- MongoDB provides several basic operations to manage and manipulate data efficiently:
  - **Create (Insert):** Add new documents to a collection.
  - **Read (Find):** Retrieve documents from a collection based on query criteria.
  - **Update:** Modify existing documents in a collection.
  - **Delete:** Remove documents from a collection.

Clients:

MongoDB Compass

- A GUI client provided by MongoDB.
- Lets you visually explore data, run queries, and manage indexes.
- Useful for developers who prefer a graphical interface over the shell

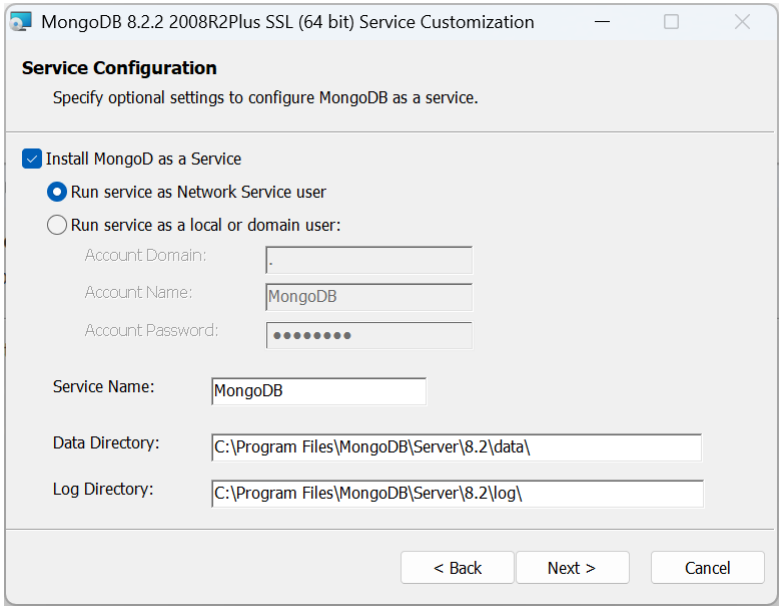
Mongo Shell (mongosh)

- Official interactive command-line client for MongoDB.
- Lets you connect to a MongoDB instance, run queries, and manage databases.
- Replaced the older mongo shell.

Install MongoDB

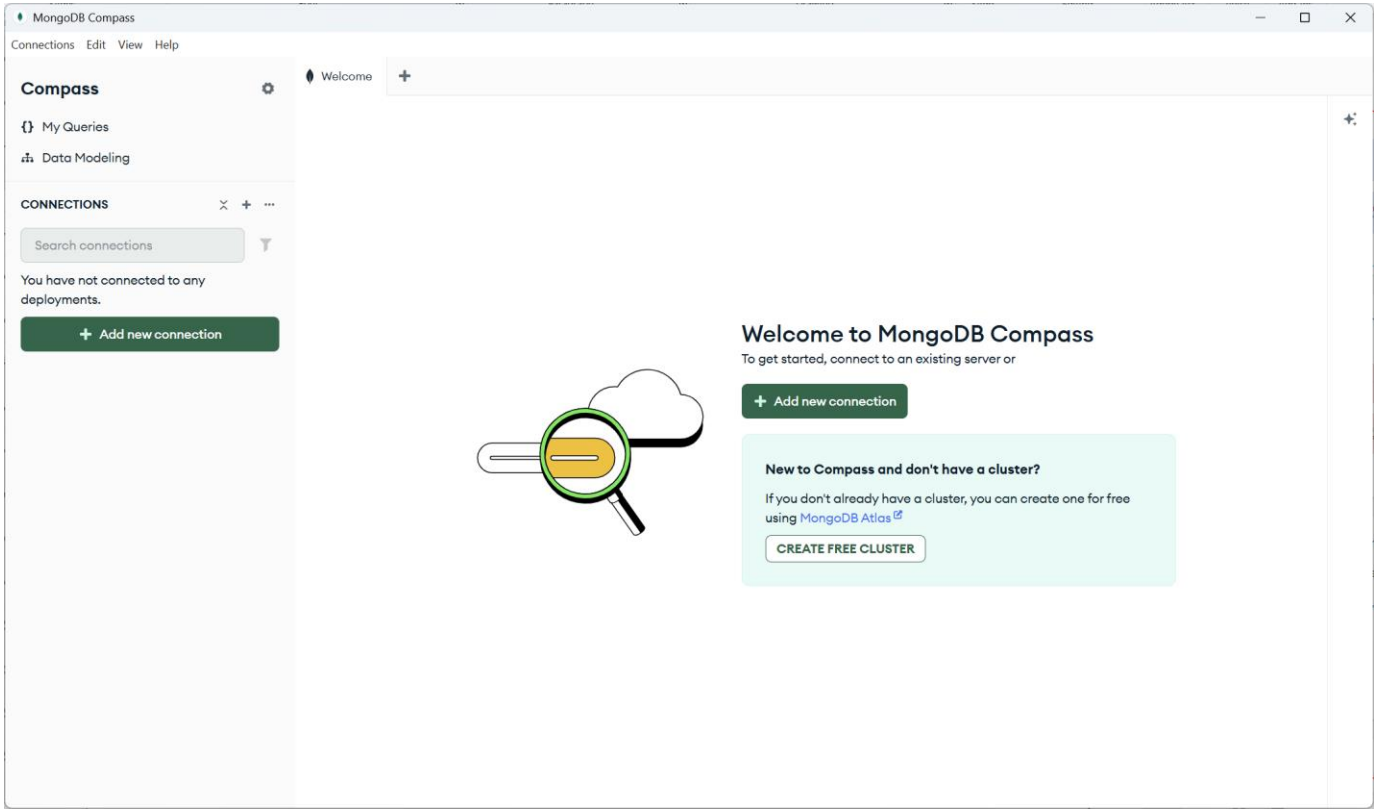
https://www.mongodb.com/try/download/community

Service Configuration:

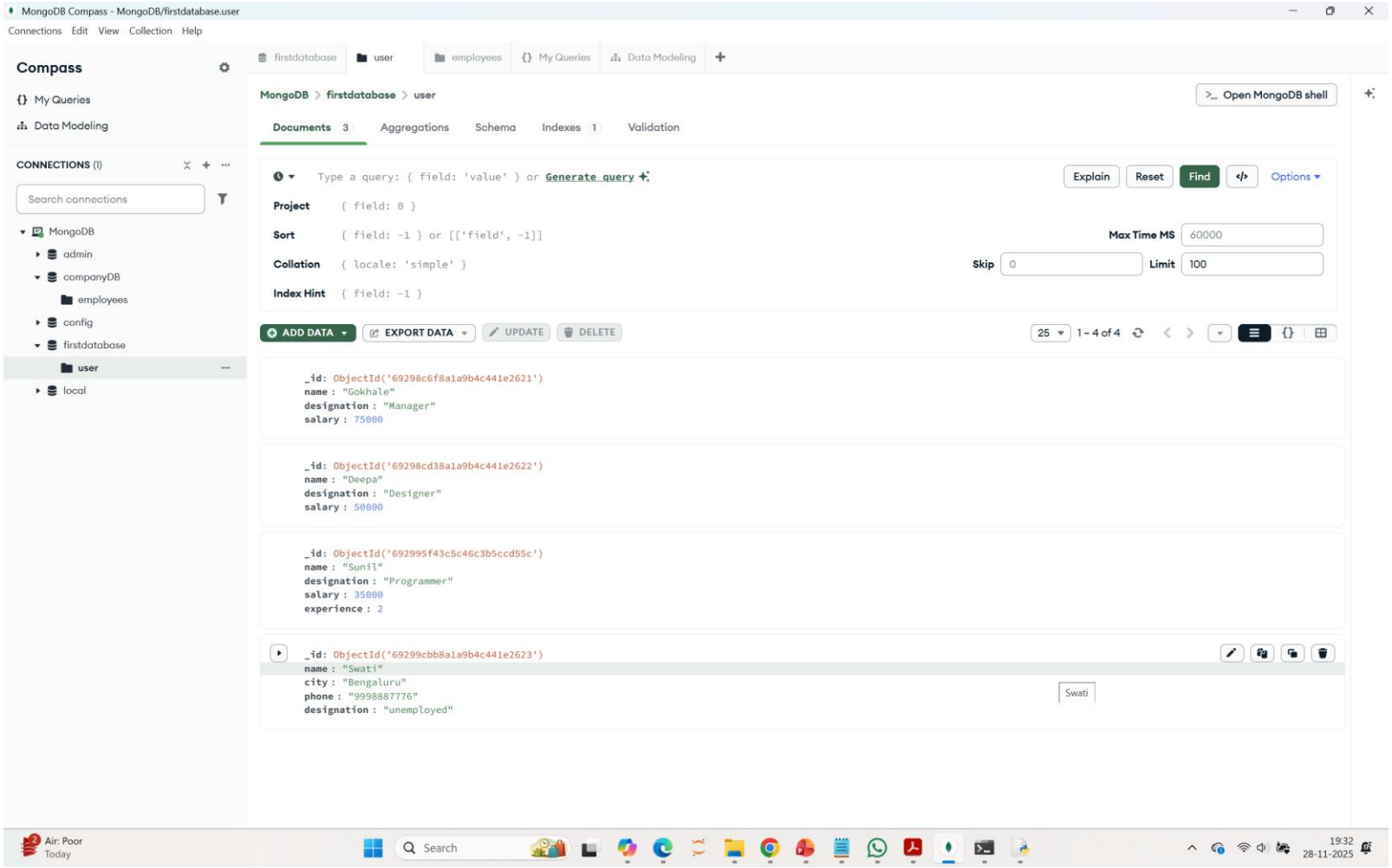


Compass

Compass: A GUI console to MongoDB to manage, administer, operate and perform other operations



Compass in Action



mongosh

CLI for performing operations

```
show dbs
use mydb
db.user.insert({name:"Gokhale", designation:"Manager" , salary: 75000})
db.user.insertOne({name:"Deepa", designation:"Designer" , salary: 50000})
db.user.find()
db.user.find({salary:50000 })
db.user.updateOne({name: 'Sunil'}, {$set:{experience: 3}})
db.user.deleteOne({name: 'Swati’})
db.user.find({ "salary": { "$gt": 35000 } })
show collections
```

Json object

```
{  "name": "Sunil",
   "designation": "Programmer",
   "salary": 35000,
   "experience": 2
}
```

Python for MongoDB

001-db-access with postgressql.py

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Topic 3: Demos of Data IO Operations

Programmatic Access to Databases

- Some Demo items

Summary

- Discussed Database structures and SQL basics for querying databases
- Understood the access methods of connecting to RDBMS using MS SQL Server & PostgreSQL
- Understood the concept of NoSQL databases with MongoDB
- Understood the Studied the working of a DV tool connecting to database

Questions?